



SEMESTER 1 EXAMINATIONS 2023/2024

MODULE MS200 - Linear Algebra

PROGRAMMES:

COMSCI	BSc in Computer Science
SE	BSc Science Education
PHA	BSc in Physics with Astronomy
AI	Chemistry with Artificial Intelligence
PBM	BSc Physics with Biomedical Sciences
PAN	BSc Physics with Data Analytics
AP	BSc in Applied Physics
CASE	BSc in Computer Applications (Sft.Eng.)

YEAR OF STUDY 2

EXAMINERS

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Dr Stephen O'Sullivan	(External)	External
Dr Ian Clancy	(External)	External

TIME ALLOWED 2 Hours

INSTRUCTIONS Attempt **ALL** 5 questions.
All questions carry equal marks.

REQUIREMENTS Approved calculators may be used.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

Question 1.

[20 marks total]

(a) Give a definition for a linear transformation $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. [4]

(b) Is the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 3$ a linear transformation? Explain your answer. [4]

(c) Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

$$f(x, y, z, w) = (2x + y - z + w, 3x + 2z - 3w, -x + 3y - 2z)$$

for all $x, y, z, w \in \mathbb{R}$. Write down the matrix representation of f . [4]

(d) Let A be the matrix representation of f from part (c). Use elementary row operations to convert A into reduced row echelon form. [8]

Question 2.

[20 marks total]

(a) Consider the system of equations

$$\begin{aligned}x + 2y - 3z &= 2, \\3x - 2y - z &= -5, \\-5x + 14y - 9z &= 7.\end{aligned}$$

Show that there is no solution to this system. [8]

(b) Consider the system of equations

$$\begin{aligned}2x - 4y - 2z - 2w &= 10, \\-x + 3y + 3z + 2w &= -2, \\-2x + 7y + 9z + 4w &= 1.\end{aligned}$$

(i) Write down the augmented matrix representing the system. [4]

(ii) Convert the augmented matrix to reduced row echelon form. [4]

(iii) Write down the form of any solution to the system. [4]

Question 3.

[20 marks total]

- (a) Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^p \rightarrow \mathbb{R}^q$ are linear transformations whose associated matrices are F and G respectively. Consider the function $k = f \circ (g + (f \circ g))$.
- (i) Find all constraints on m , n , p and q which must hold in order for k to be well-defined. [3]
 - (ii) What is the domain and codomain of k ? [2]
 - (iii) Let K be the matrix representation of k . What are the dimensions of K ? [1]
 - (iv) Write K in terms of F and G . [2]
- (b) Use the method of least squares to find the best straight line fit to the set of data points $\{(1, 4), (2, 2), (3, 5), (4, 8)\}$. [12]

Question 4.

[20 marks total]

Consider the 3×3 matrix

$$A = \begin{pmatrix} 1 & 4 & -5 \\ -2 & -6 & 6 \\ 3 & 10 & -12 \end{pmatrix}.$$

- (a) Find $\det(A)$, the determinant of A . [5]
- (b) Find A^{-1} , the inverse of A . [7]
- (c) Suppose that B is a 3×3 matrix and $\det(AB) = 6$. Find $\det(B)$? [4]
- (d) Using your answer from the previous part, if the eigenvalues of B are $\lambda_1 = 1$ and $\lambda_2 = -1$, and λ_3 , what is the value of λ_3 ? [4]

Question 5.

[20 marks total]

Let

$$A = \begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix}.$$

- (a) Find the eigenvalues of A . [5]
- (b) Find eigenvectors for each eigenvalue of A . [5]
- (c) Express A as a product EDE^{-1} where D is a diagonal matrix. [6]
- (d) Using your answer to part (c), find A^{10} . [4]